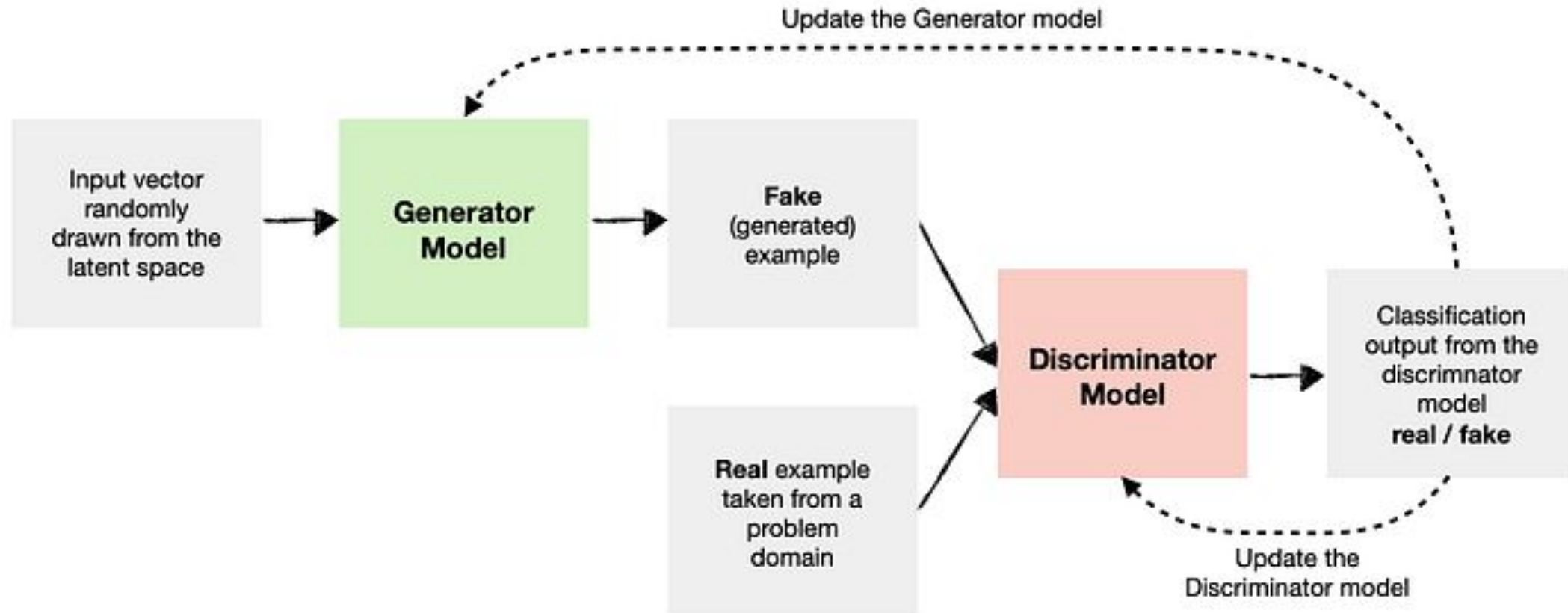


Vanilla GAN in a NutShell(Recap)

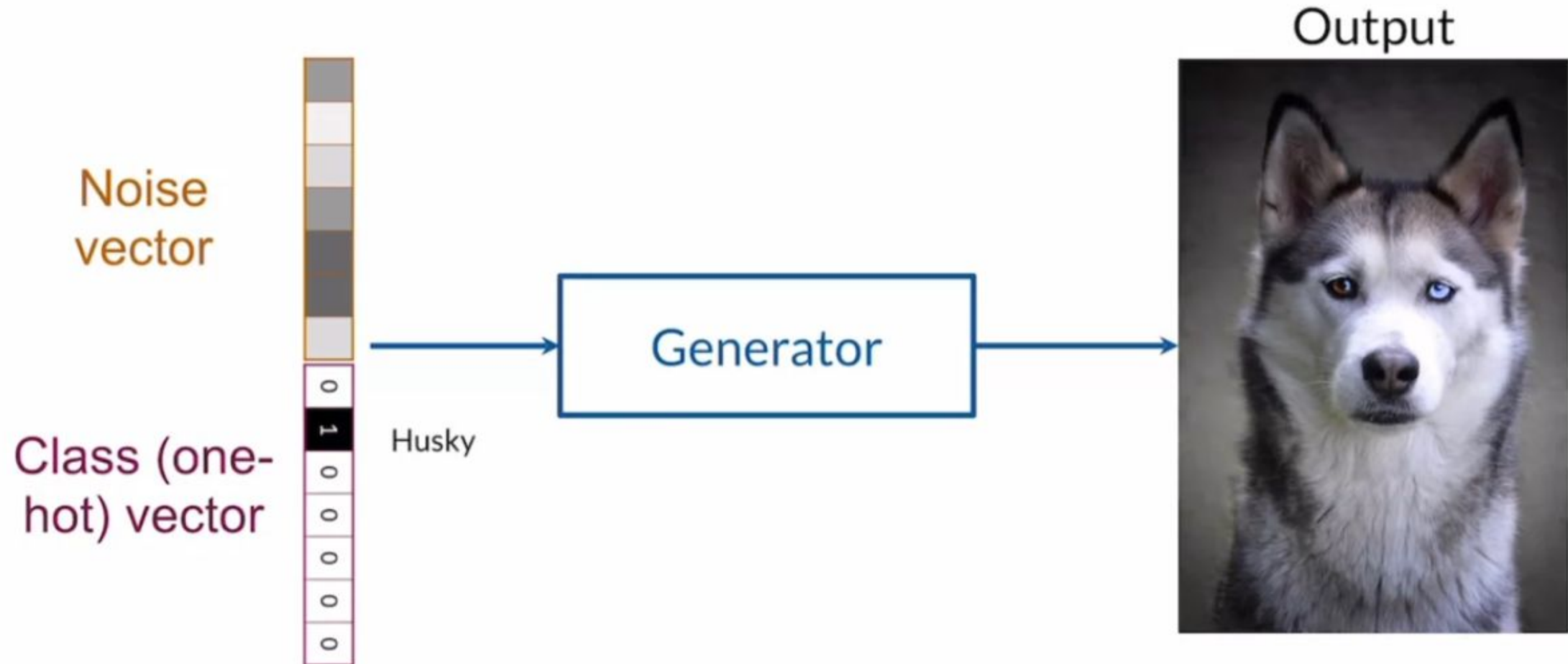


$$\min_G \max_D V(D, G)$$

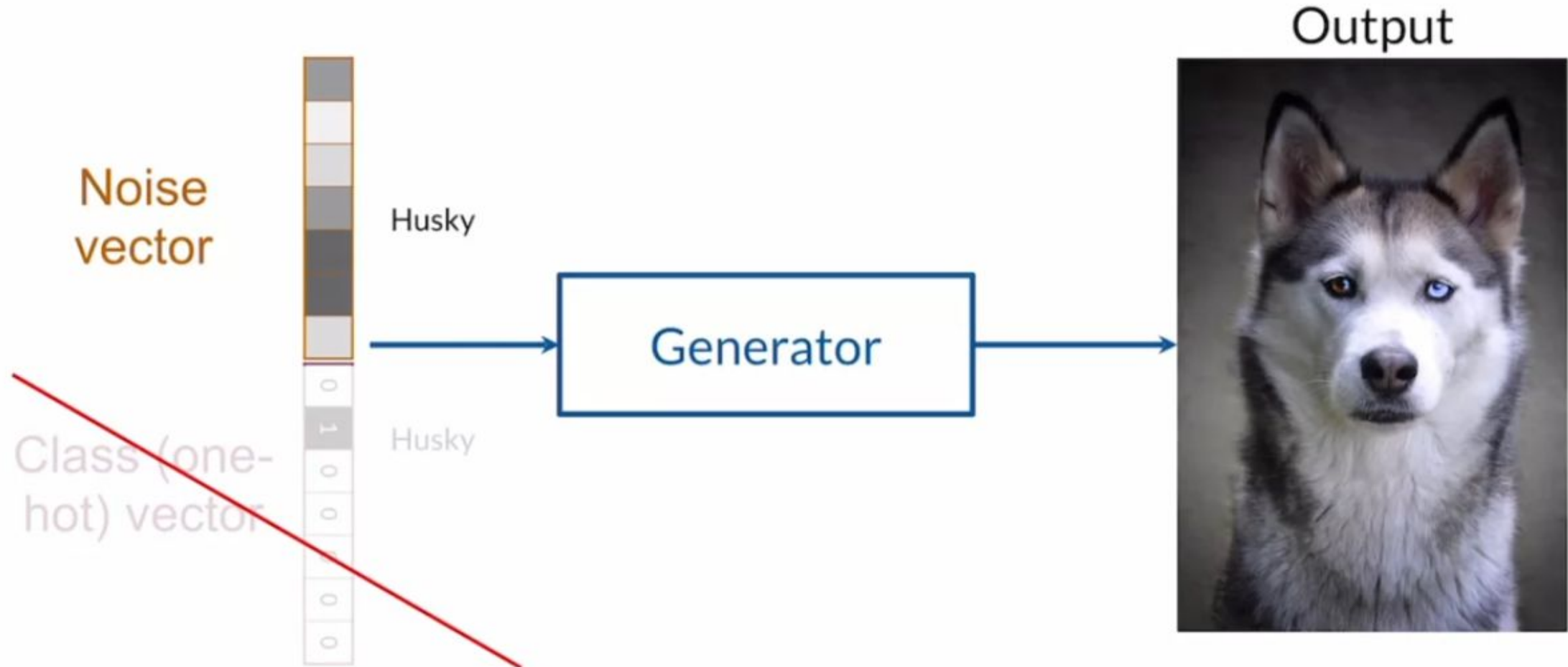
- It is formulated as a **minimax game**, where:
 - The Discriminator is trying to maximize its reward $V(D, G)$
 - The Generator is trying to minimize Discriminator's reward (or maximize its loss)

$$V(D, G) = \mathbb{E}_{x \sim p(x)} [\log D(x)] + \mathbb{E}_{z \sim q(z)} [\log(1 - D(G(z)))]$$

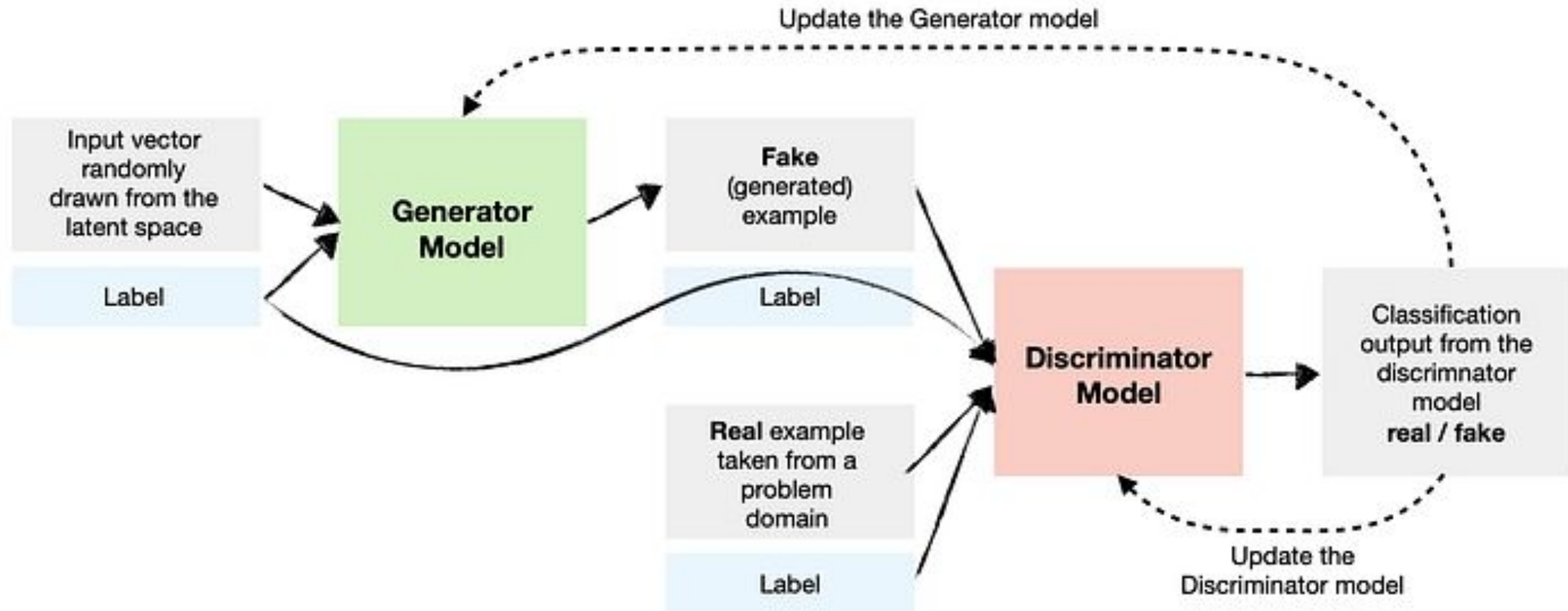
Encourage Disentanglement: Supervision



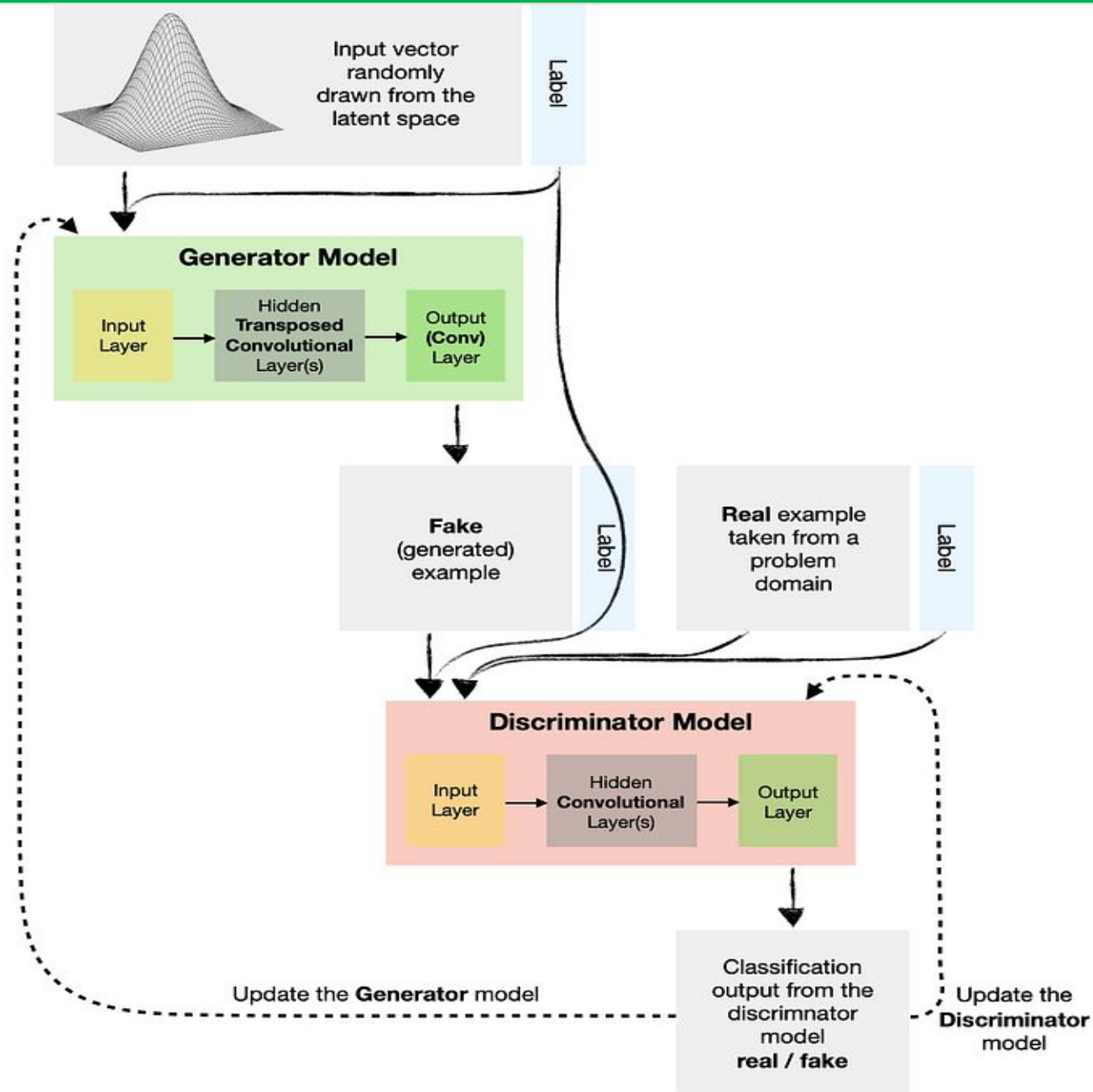
Encourage Disentanglement: Supervision



Conditional GAN(cGan)

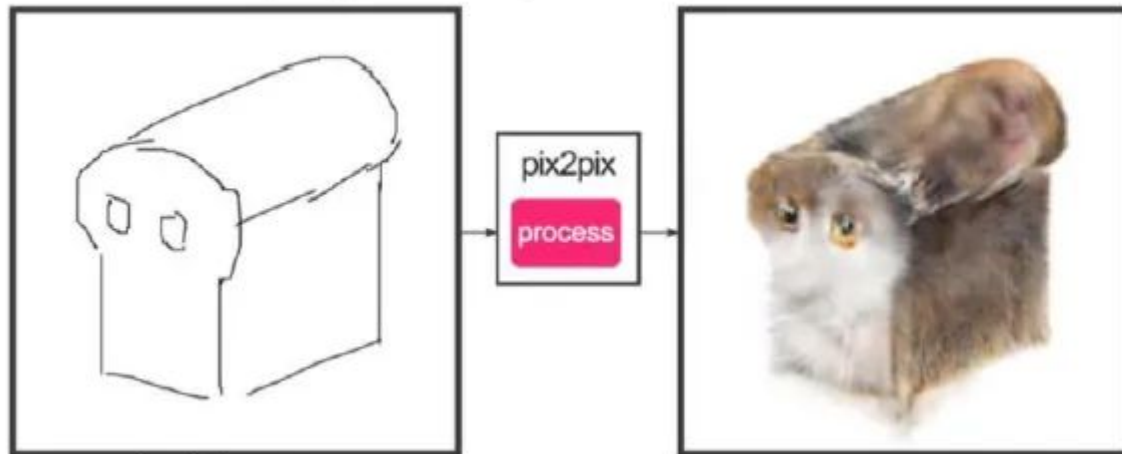


Conditional GAN(cGan)



Conditional GAN Examples

#edges2cats by Christopher Hesse



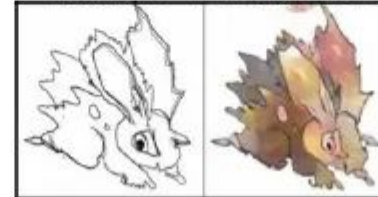
sketch by Ivy Tsai

Background removal



by Kaihu Chen

Sketch → Pokemon



by Bertrand Gondouin

Palette generation



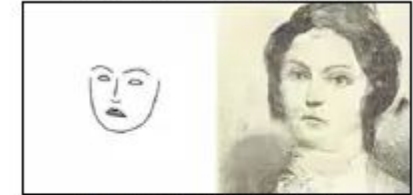
by Jack Qiao

"Do as I do"



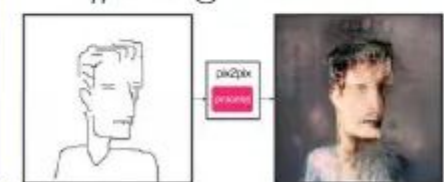
by Brannon Dorsey

Sketch → Portrait



by Mario Klingemann

#fotogenerator



sketch by Yann LeCun

Image-to-Image Translation

Edges to photo



Paired
generation

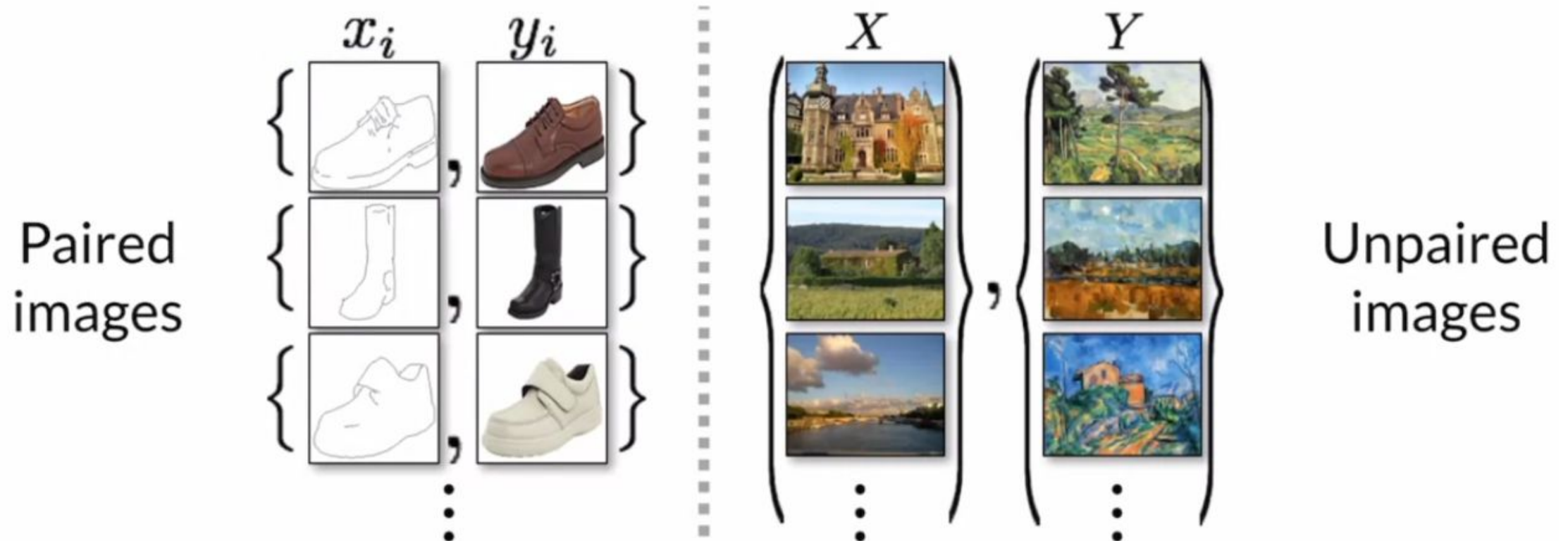
Monet to photo



Unpaired
generation

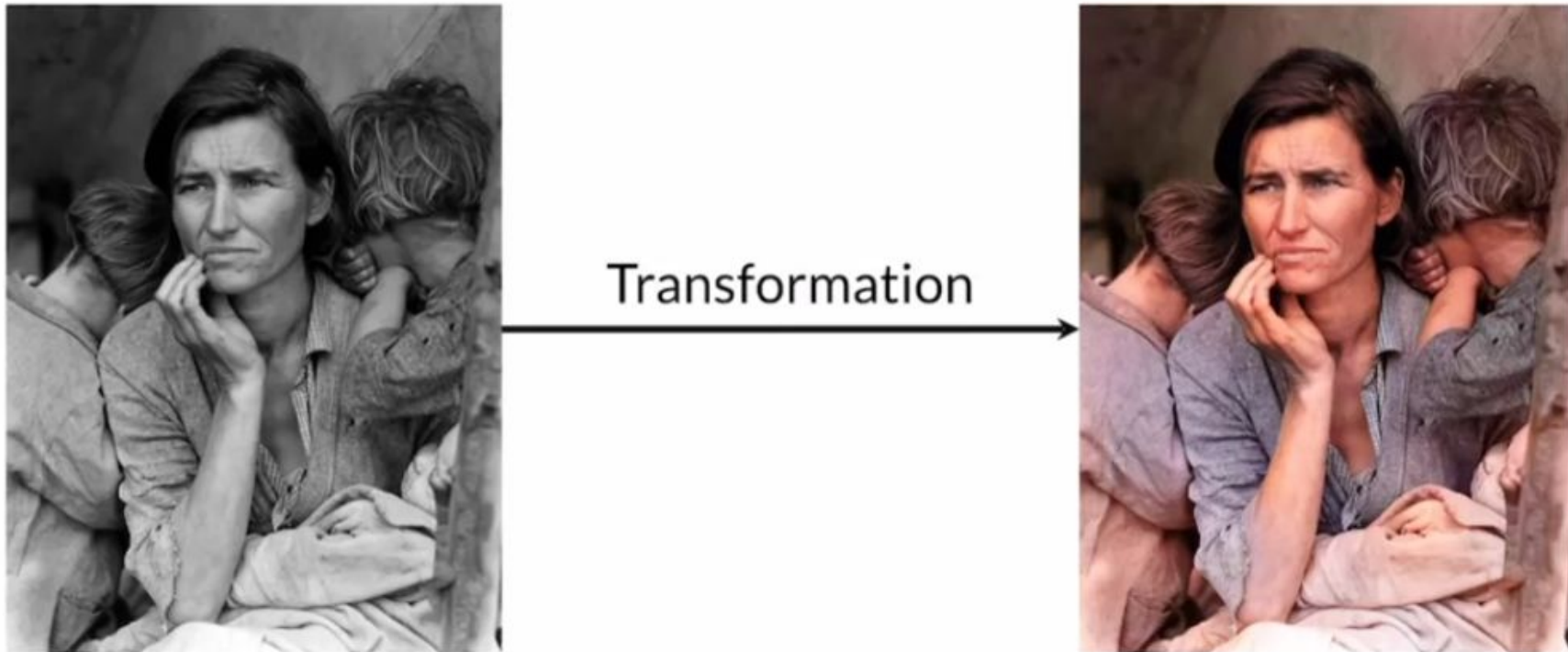
Available from: <https://arxiv.org/abs/1611.07004>

Image-to-Image Translation



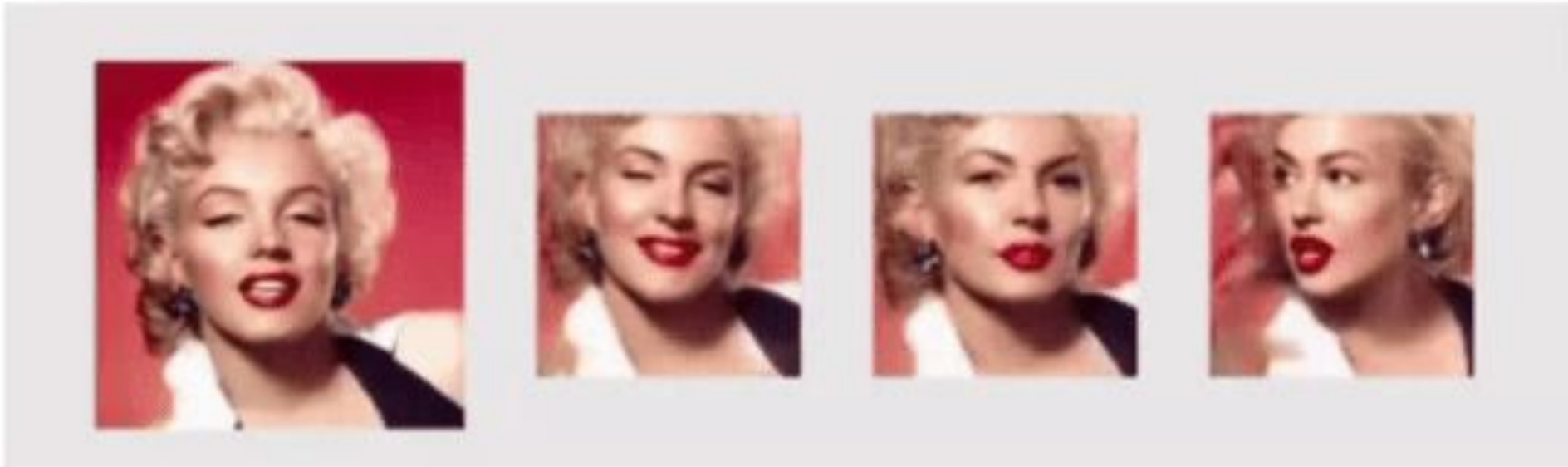
Available from: <https://arxiv.org/abs/1703.10593>

Image-to-Image Translation



Available from: <https://twitter.com/citnaj/status/1124904251128406016>

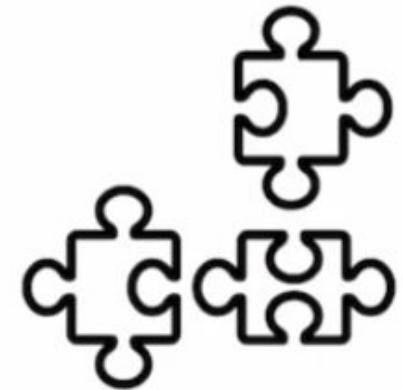
Neural Talking Head



Paired Image-to-Image Translation



- U-Net generator: image $\rightarrow$ image
- PatchGAN discriminator
 - Inputs input image and paired output (either real target or fake)
 - Outputs classification matrix
- Generator loss has a regularization term



Pix2Pix

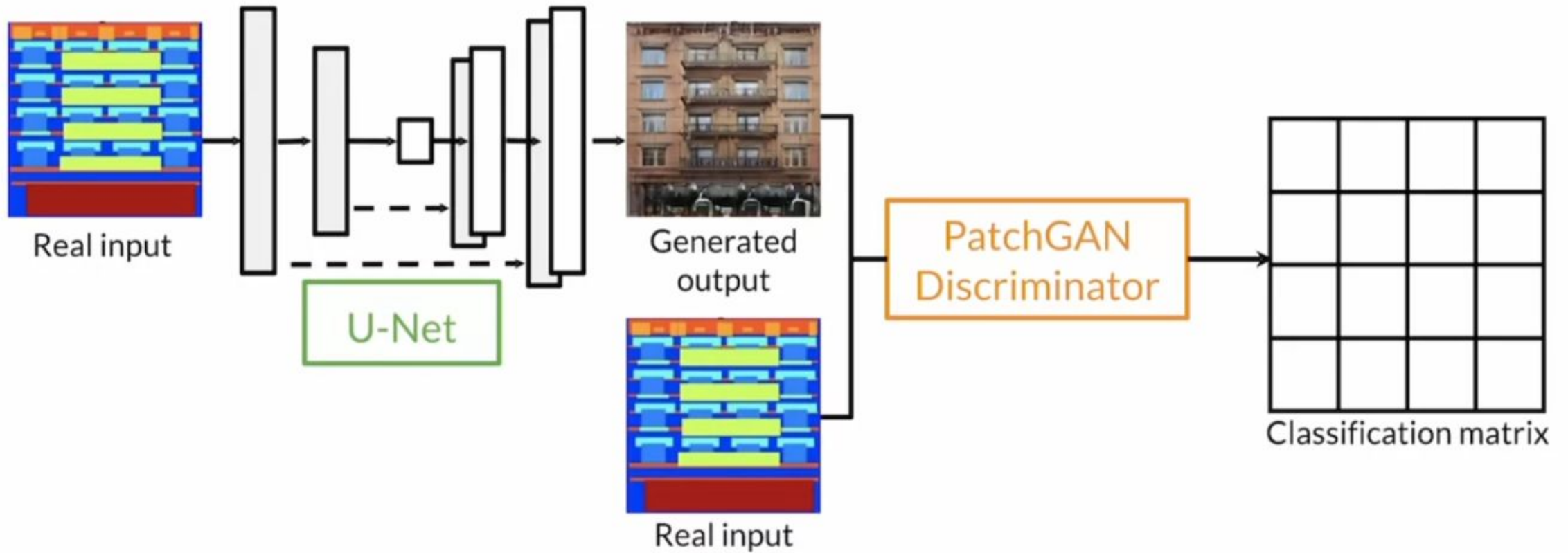


Image available from: <https://arxiv.org/abs/1611.07004>

Pix2Pix: Discriminator Loss

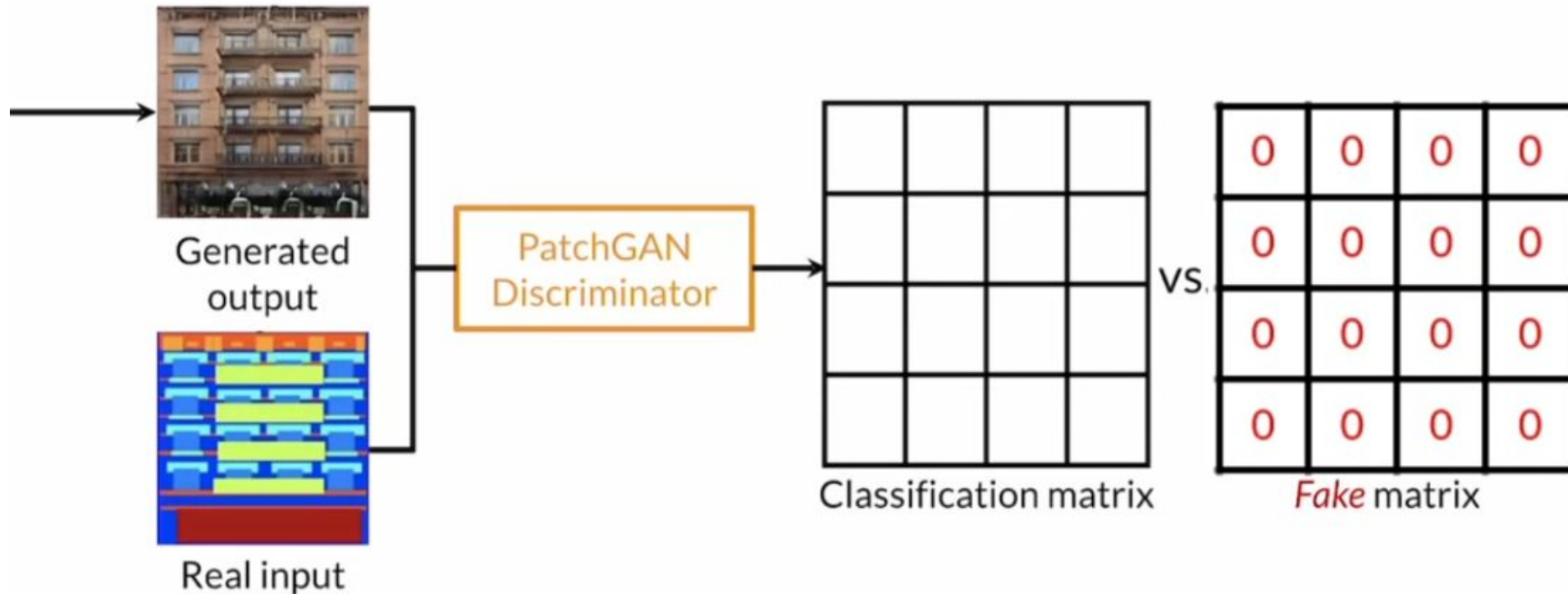


Image available from: <https://arxiv.org/abs/1611.07004>

Pix2Pix: Discriminator Loss

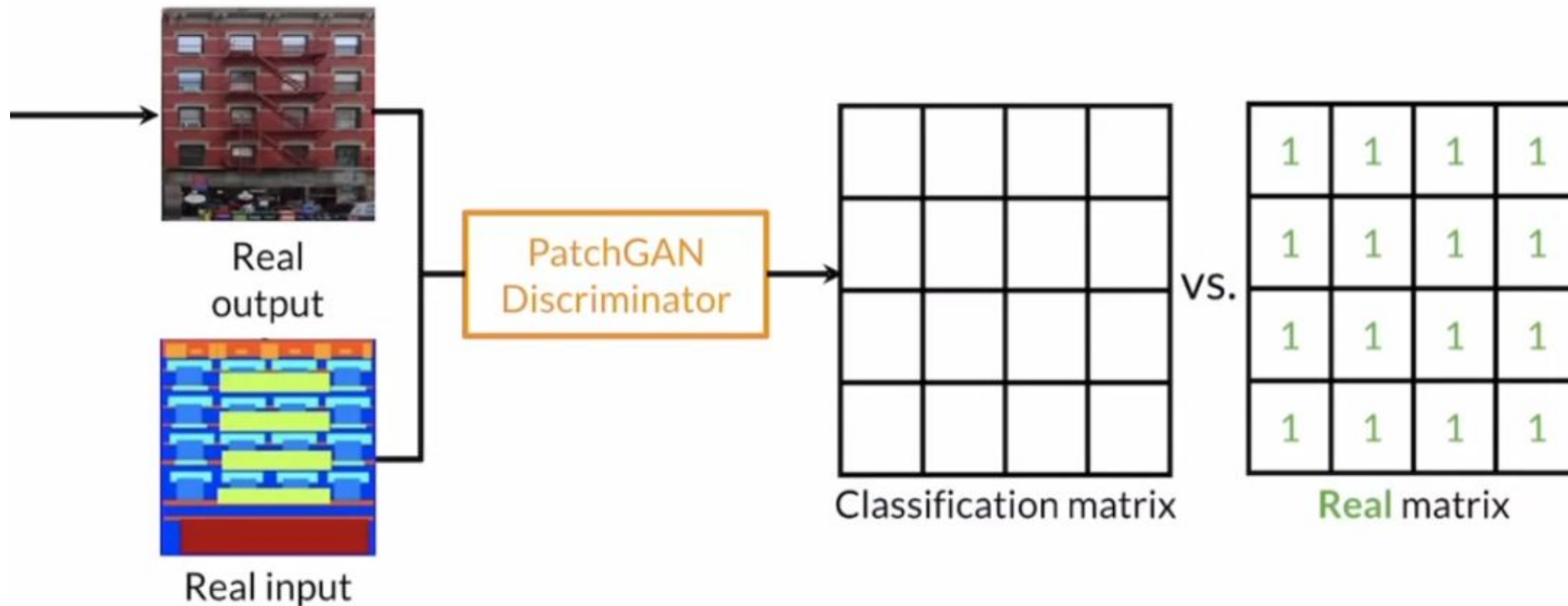


Image available from: <https://arxiv.org/abs/1611.07004>

Pix2Pix: Generator Loss

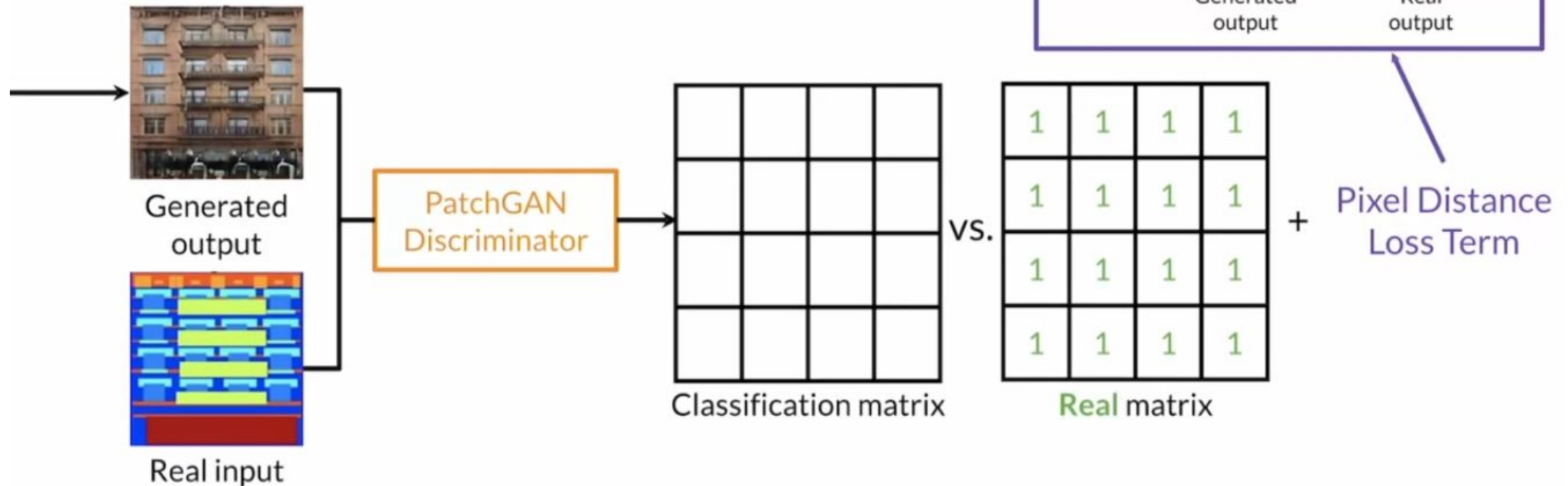


Image available from: <https://arxiv.org/abs/1611.07004>

Pix2Pix Discriminator: PatchGAN

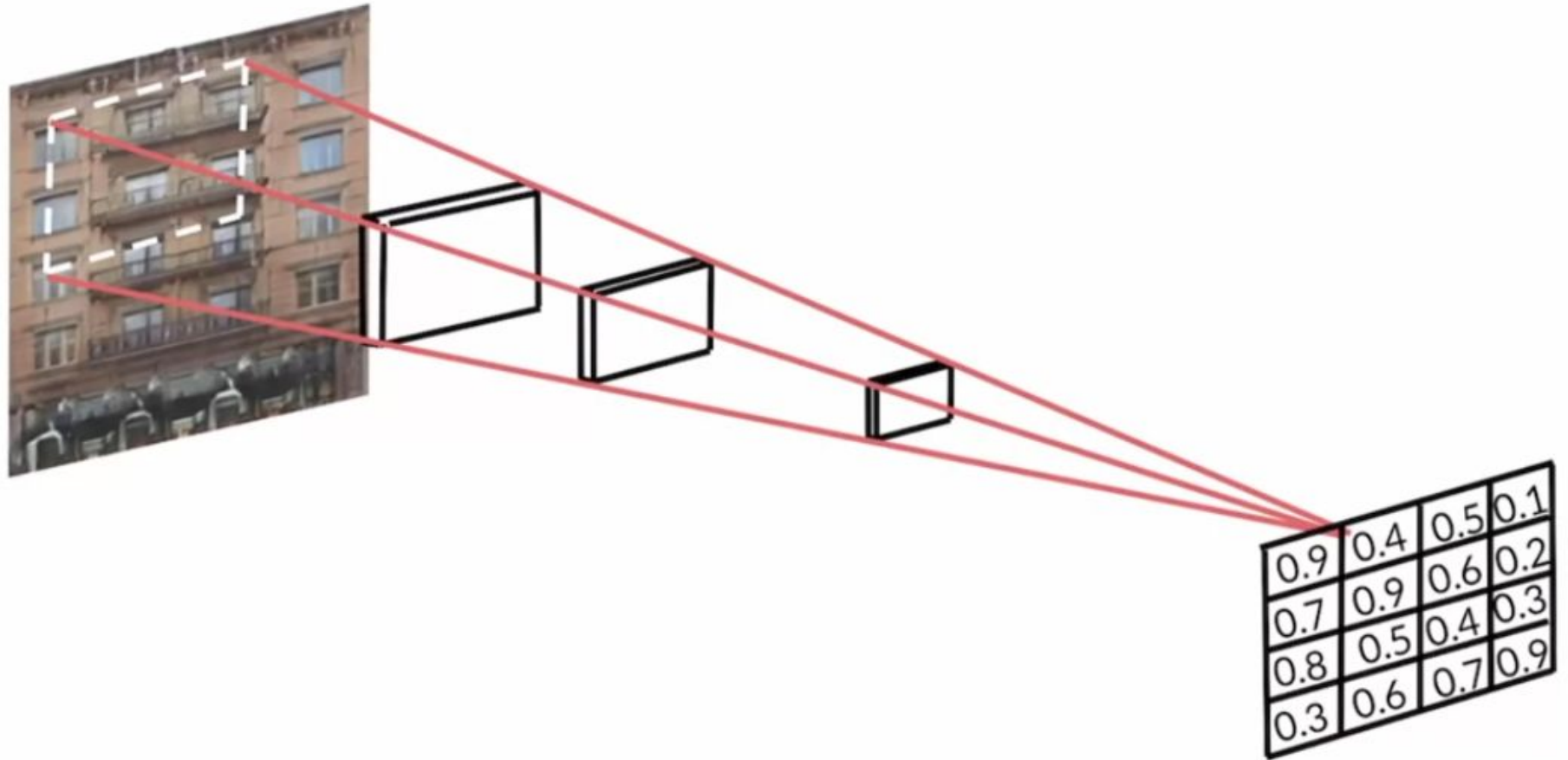
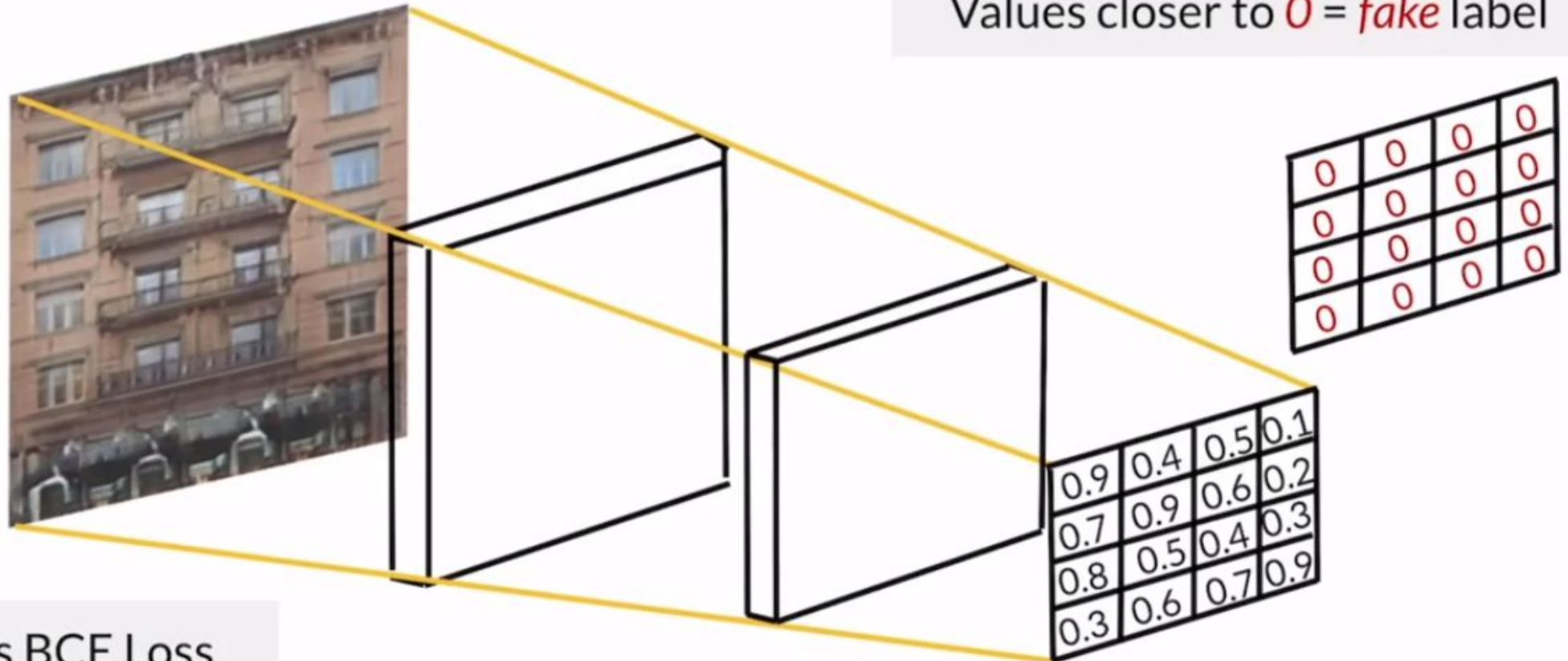


Image available from: <https://arxiv.org/abs/1611.07004>

Based on: <https://arxiv.org/abs/1803.07422>

Pix2Pix Discriminator: PatchGAN

Values closer to *0* = *fake* label



Still uses BCE Loss

Image available from: <https://arxiv.org/abs/1611.07004>
Based on: <https://arxiv.org/abs/1803.07422>

- PatchGAN discriminator outputs a matrix of values, each between 0 and 1
- Label matrices:
 - 0's = fake
 - 1's = real



Unpaired Image-to-Image Translation



Photograph



Monet



Van Gogh



Cezanne



Ukiyo-e

Available from: <https://arxiv.org/abs/1703.10593>

Unpaired Image-to-Image Translation

- Unpaired image-to-image translation:
 - Learns a mapping between two piles of images
 - Examines common elements of the two piles (content) and unique elements of each pile (style)
- Unlike paired image-to-image translation, this method is unsupervised

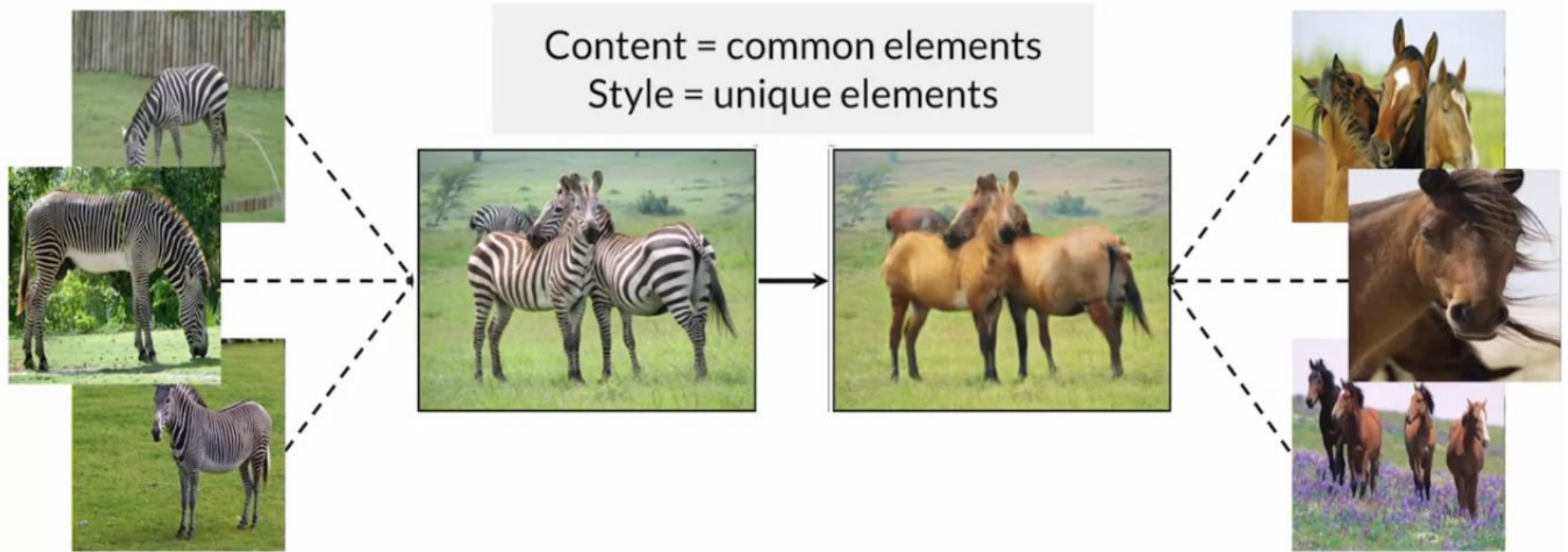


- CycleGAN has four components:
 - Two generators
 - Two discriminators
- The inputs to the generators and discriminators are similar to Pix2Pix, except:
 - There are no real target outputs
 - Each discriminator is in charge of one pile of images



Unpaired Image-to-Image Translation

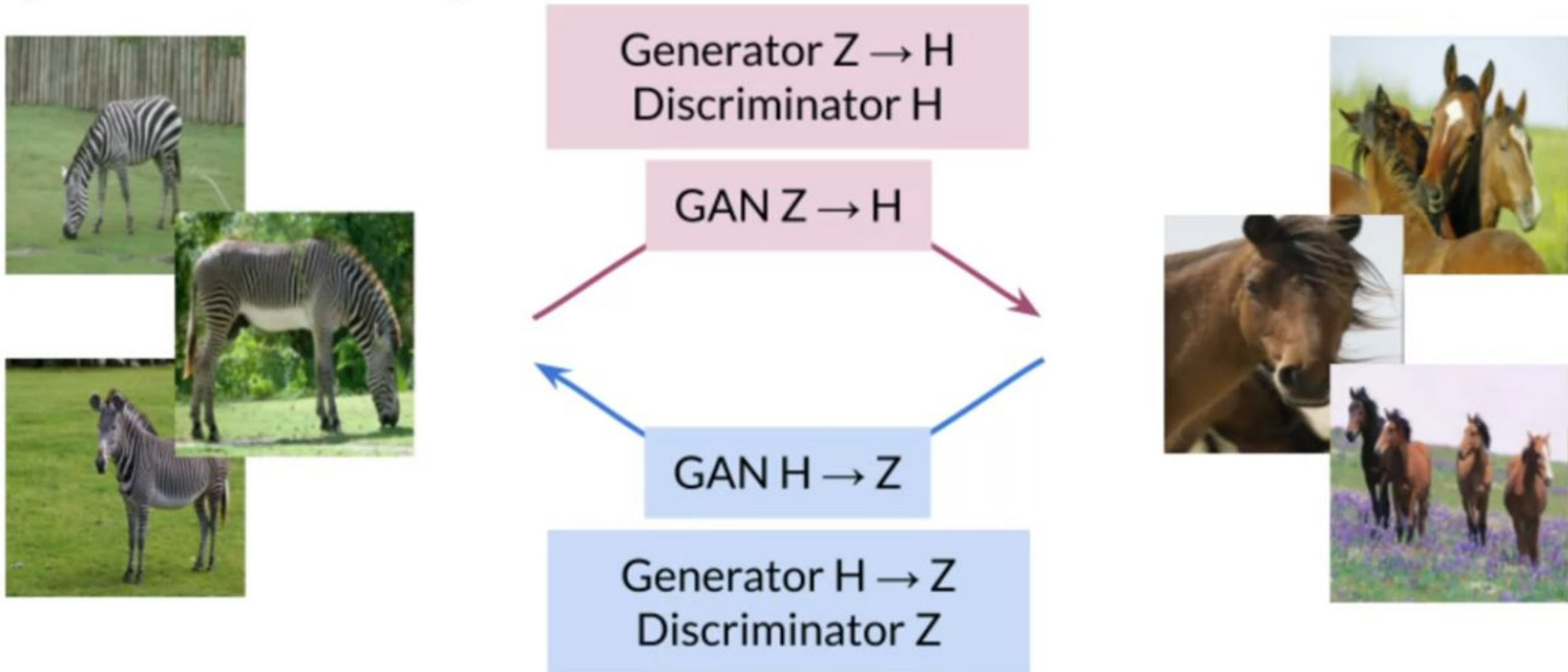
Mapping Between Two Piles



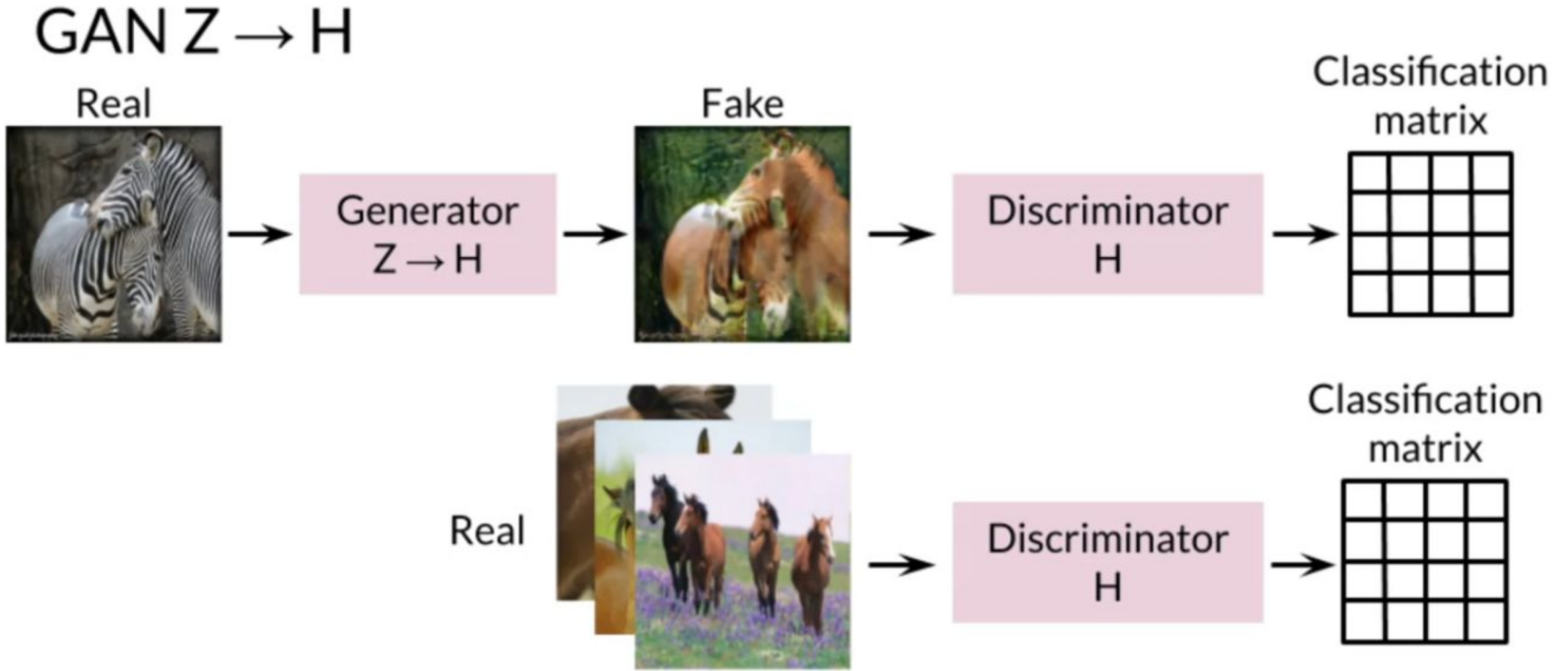
(Center) Images available from: <https://arxiv.org/abs/1703.10593>

(Sides) Images available from: <https://github.com/togheppi/CycleGAN>

CycleGAN Components

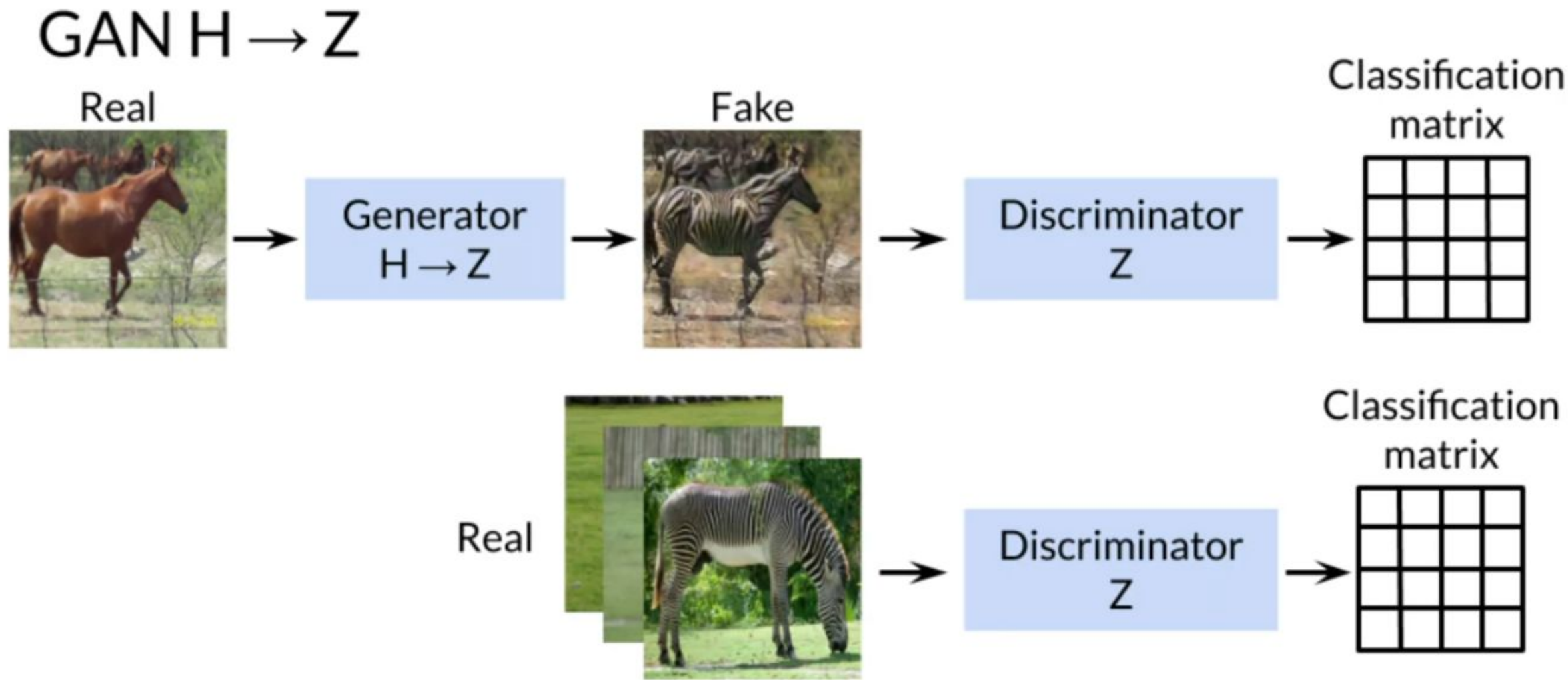


Images available from: <https://github.com/togheppi/CycleGAN>



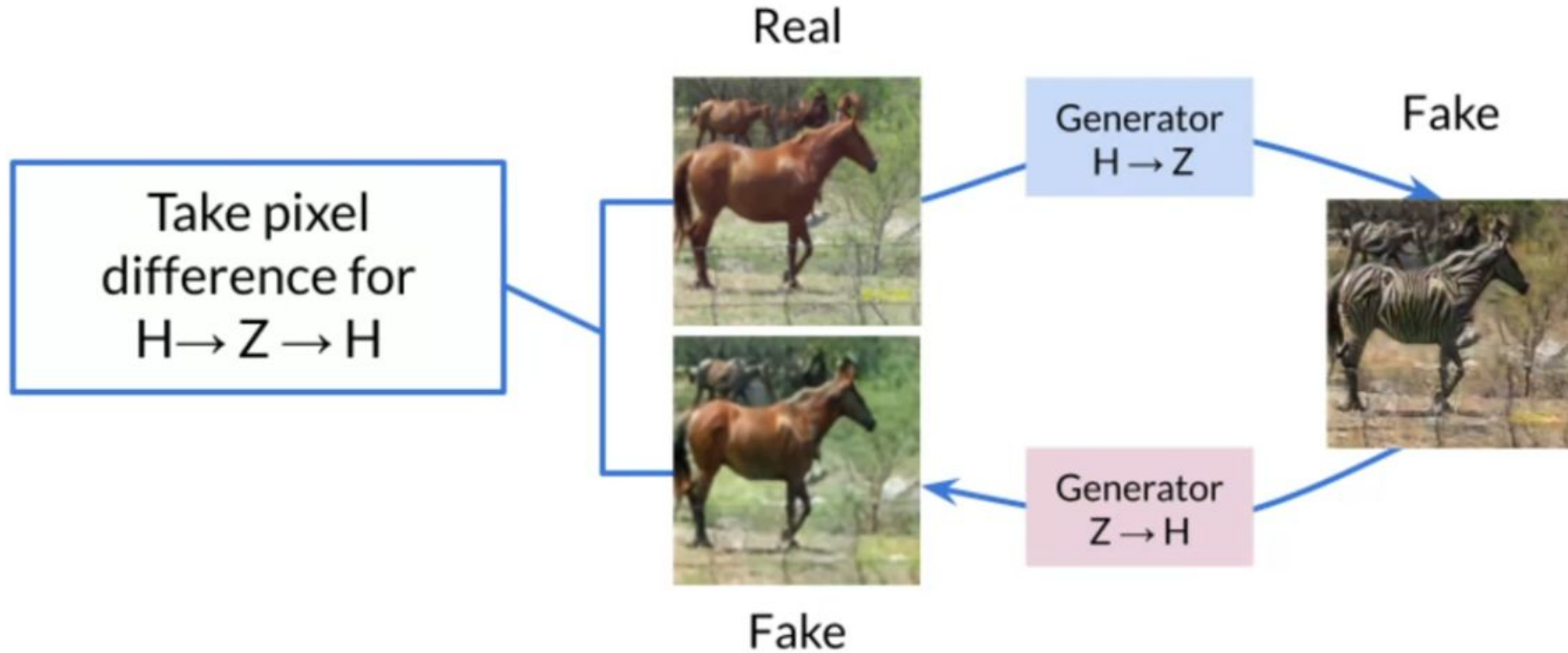
Images available from: <https://github.com/togheppi/CycleGAN>

CycleGAN



Images available from: <https://github.com/togheppi/CycleGAN>

Cycle Consistency Loss



Images available from: <https://github.com/togheppi/CycleGAN>

Cycle Consistency Loss

$$\sum_i \left| \text{Image}_1 - \text{Image}_2 \right| + \sum_i \left| \text{Image}_3 - \text{Image}_4 \right|$$

$Z \rightarrow H \rightarrow Z$ $H \rightarrow Z \rightarrow H$

Cycle Consistency Loss is the sum of both directions

Images available from: <https://github.com/togheppi/CycleGAN>

Cycle Consistency Loss

Adversarial Loss

+

$$\sum_i | \text{img}_1 - \text{img}_2 | + \sum_i | \text{img}_3 - \text{img}_4 |$$

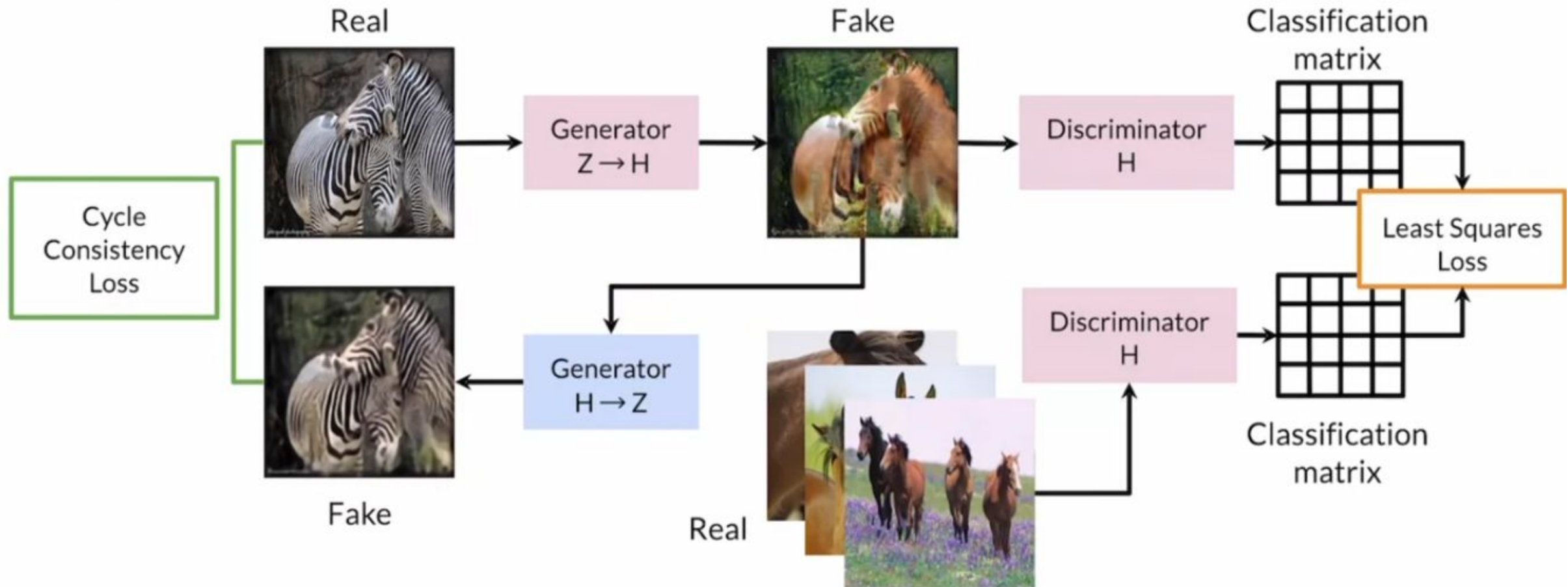
$Z \rightarrow H \rightarrow Z$ $H \rightarrow Z \rightarrow H$

Images available from: <https://github.com/togheppi/CycleGAN>

Cycle Consistency Loss

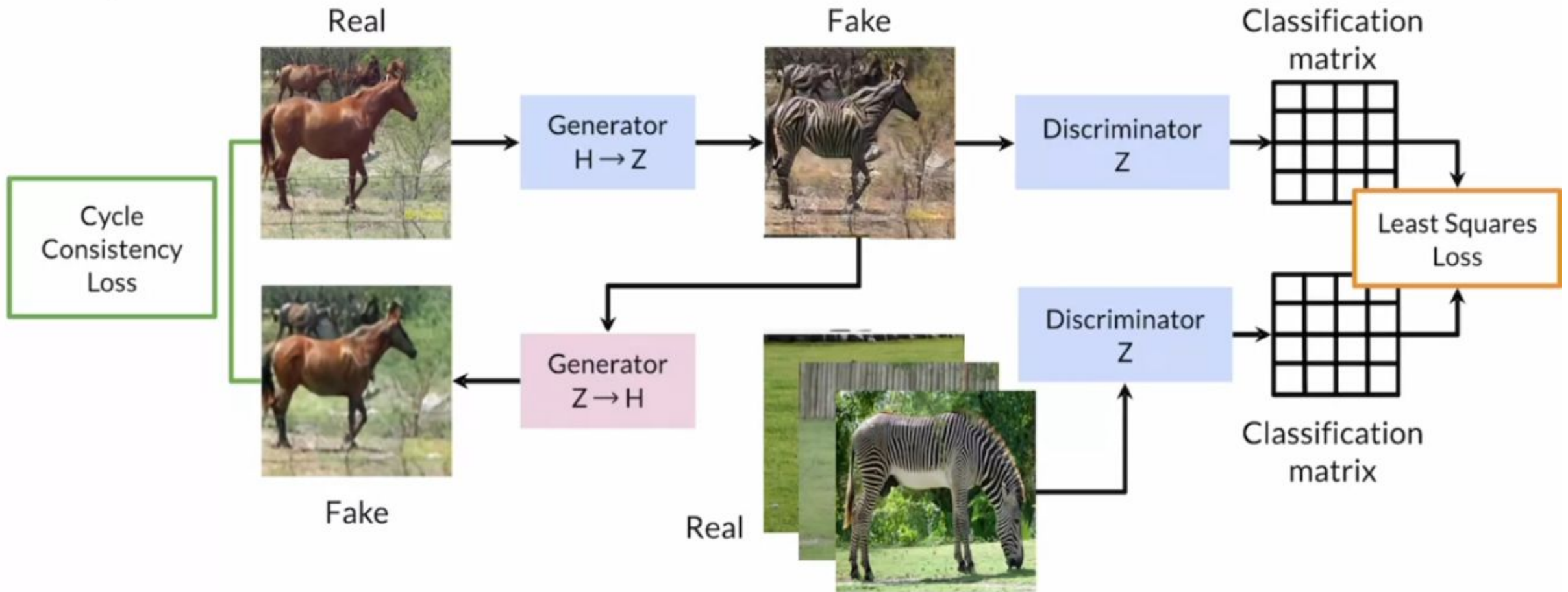
Adversarial Loss + λ * Cycle Consistency Loss

CycleGAN



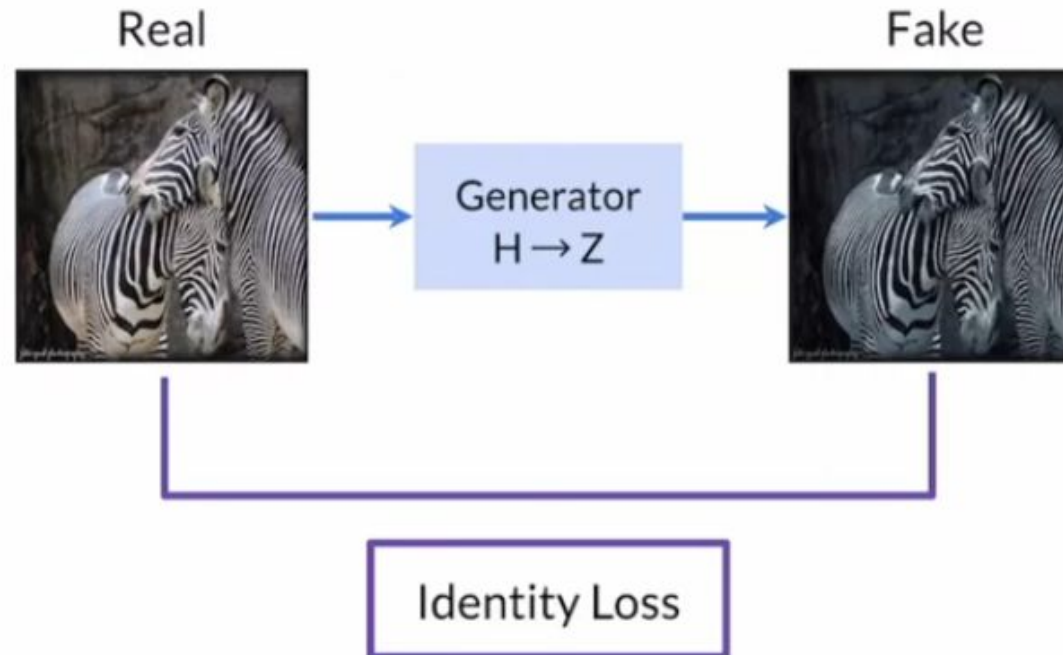
Images available from: <https://github.com/togheppi/CycleGAN>

CycleGAN



Images available from: <https://github.com/togheppi/CycleGAN>

CycleGAN



Images available from: <https://github.com/togheppi/CycleGAN>

CycleGAN Loss

$$\begin{aligned} \text{Generator Loss} &= \text{Least Squares Adversarial Loss } H \rightarrow Z + \text{Cycle Consistency Loss } H \rightarrow Z \rightarrow H + \text{Identity Loss } H \rightarrow Z \\ &+ \text{Least Squares Adversarial Loss } Z \rightarrow H + \text{Cycle Consistency Loss } Z \rightarrow H \rightarrow Z + \text{Identity Loss } Z \rightarrow H \end{aligned}$$